



SECURITY VERIFICATION REPORT

CMTAT

Automated symbolic-execution-driven fuzz testing

A negative-assurance fuzzing engagement against **CMTA/CMTAT** — Olympix enumerated execution paths across the 3 highest-value contracts, generated concrete adversarial Foundry tests under 0 attack strategies, and ran every one — with a runnable artifact behind every number.

PATHS EXPLORED	FEASIBLE	TEST CASES	FILES	EXPLOITS
35,188	27,252	0	0	0

Contents

01 Executive summary

02 Why this matters

03 How the fuzzing engine works

04 Fuzz testing results

01 Executive summary

Olympix ran its symbolic-execution-driven fuzzing engine against the CMTAT protocol, probing the core state-changing surface with concrete adversarial execution.

35,188

execution paths
enumerated across the 3
highest-value contracts

0

successful exploits across 0
adversarial test cases & 0
attack strategies

0

adversarial test cases
generated & executed

0

runnable Foundry test files
produced

Symbolic-execution-driven fuzzing. Explored 35,188 execution paths across the 3 highest-value contracts, kept the 27,252 feasible (satisfiable) ones, generated and ran 0 targeted test cases under 0 adversarial strategies, and produced **zero successful exploits** — a strong negative-assurance result for the core surface.

Fuzz testing

35,188 paths · 27,252 feasible · 0 test cases · 0 generated Foundry files

0 exploits

across all 0 attack strategies

Engagement at a glance

Target	CMTA/CMTAT
Commit	49544f4de1993008acfc9e848d0bf03bd31d8579
Branch	master
Scope	3 contract(s) deep-fuzzed · 35,188 paths enumerated
Toolchain	Olympix Fuzz Test Generator (Foundry + symbolic execution)
Run	session f37cd9fe-cb1e-4732-821a-03c6811416dc · 6 minutes wall-clock · July 16, 2026 UTC

Maturity note — the fuzz engine is a work in progress

The fuzzing engine is under active development. The 0 adversarial strategies exercised in this run are the engine's current set; additional strategies and deeper multi-call sequences land with each release, so today's result is a floor on coverage, not a ceiling.

02 Why this matters

Negative assurance you can show your community

“We fuzzed it” is easy to claim. 35,188 symbolically-enumerated paths, 0 concrete adversarial test cases and 0 successful exploits is a claim with artifacts behind every number — every test case is a runnable Foundry file in this package.

Real exploit attempts, not warnings

Each generated test is a genuine attack — funding the contract, pranking an attacker, deploying real reentrancy harnesses — whose assertions fail if the attack succeeds. A clean run means the attacks were tried and held.

Path-aware, not blind

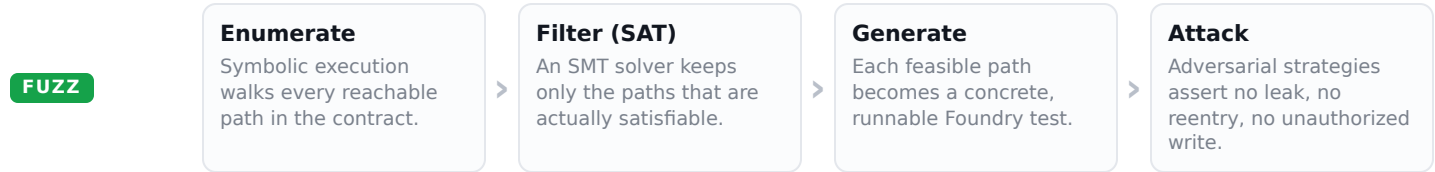
Symbolic execution enumerates reachable paths and an SMT solver keeps only the feasible ones, so the adversarial tests concentrate on inputs that can actually occur rather than random noise.

Fast enough for every release

Specification to result took 6 minutes. This can run on every release candidate, not once per audit cycle.

03 How the fuzzing engine works

The engine starts from whole-contract symbolic execution, turns feasible paths into concrete adversarial tests, and runs every one.



Fuzzing executes thousands of concrete adversarial transactions and shows what attackers actually achieve against the contract under realistic conditions.

04 Fuzz testing results

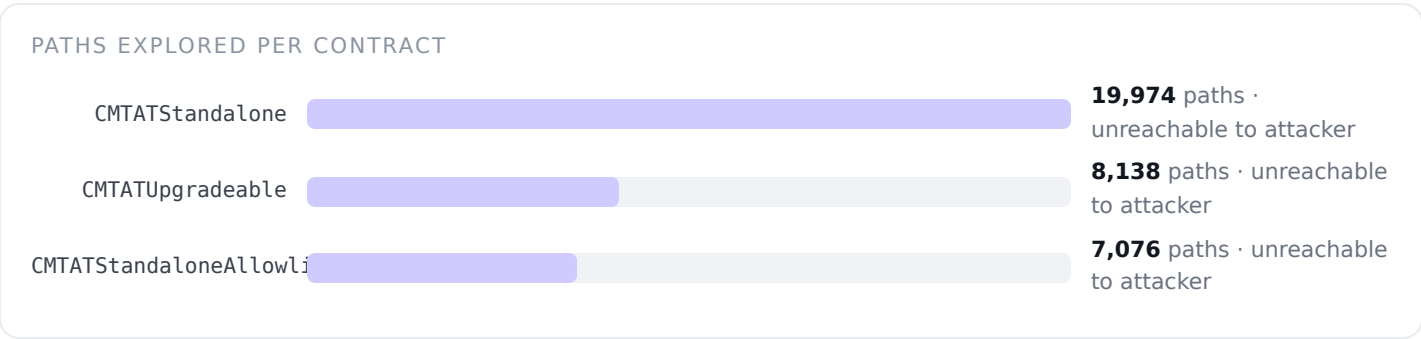
For each contract, Olympix enumerated execution paths, kept the feasible ones, and generated concrete Foundry tests that drive every feasible path under each adversarial strategy. Each test is a real exploit attempt whose assertions fail if the attack succeeds.



Results by contract

Contract	Paths	Feasible	Test cases	Files	Exploits
CMTATStandalone	19,974	15,518	0	0	0
CMTATUpgradeable	8,138	5,910	0	0	0
CMTATStandaloneAllowlist	7,076	5,824	0	0	0
Total	35,188	27,252	0	0	0

Contracts with no attacker-reachable test cases produced zero satisfiable paths under attacker preconditions — their guarded entry points are simply unreachable to an unauthorized caller. These are positive signals, not coverage gaps.



Proactive Web3 smart contract security

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